

H1 Statistics — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

H1_Q1: Data Collection and Types

Hook

A school wants to find out which lunch option is most popular among students. Should they ask every single student (2000 of them), or just ask one class? How should they record the answers — as 'Chicken', 'Pasta', 'Veggie' or as numbers? The choices you make about collecting and classifying data determine whether you get useful results or misleading nonsense.

Prerequisites

- Basic arithmetic (counting and simple percentages)
- Understanding of sorting and grouping items into categories
- Familiarity with tally marks

Core Explanation

Data is information collected for analysis. How you collect and classify it matters.

Primary vs Secondary Data

- **Primary data** — collected directly by you (surveys, experiments, observations). Fresh and specific, but time-consuming and costly.
- **Secondary data** — already collected by someone else (government stats, published reports, websites). Quick and cheap, but may not perfectly fit your needs.

Qualitative vs Quantitative Data

- **Qualitative (categorical)** — describes qualities or categories (e.g., eye colour, favourite subject, type of car). Cannot be measured numerically.
- **Quantitative (numerical)** — involves numbers and measurements (e.g., height in cm, number of siblings, test score). Can be further split.

Discrete vs Continuous Data

- **Discrete data** — can only take specific, separate values (usually whole numbers). Counts of things: number of students in a class, shoe sizes, number of cars passing.
- **Continuous data** — can take any value within a range. Measurements: height, weight, time, temperature.

Data Collection Methods

- **Surveys / Questionnaires** — ask people. Must avoid biased wording.
- **Experiments** — controlled conditions to measure outcomes.
- **Observations** — watch and record without interfering.
- **Census** — survey every member of a population (accurate but expensive).
- **Sample** — survey a subset (cheaper, faster, but can have sampling bias).

****Tally Charts****

A tally chart is a quick way to record data as it is collected. Groups of 5 are typically drawn as four vertical lines with a diagonal cross ($\cancel{\{\}}\{\}$). This makes counting totals easy.

Worked Example

A student asks 20 classmates: 'How many siblings do you have?' The raw answers are: 2, 1, 0, 3, 1, 2, 1, 0, 2, 1, 4, 1, 2, 0, 1, 2, 1, 3, 2, 1

****Step 1 — Organise with a tally chart****

Number of siblings	Tally	Frequency
0	III	3
1	IIII	7
2	IIII	6
3	II	2
4	I	1
5+		0

****Step 2 — Classify the data****

- Type of data: ****Quantitative**** (numbers about siblings)
- Sub-type: ****Discrete**** (you can't have 2.5 siblings)
- Collection method: ****Primary data**** (the student asked classmates themselves)

****Step 3 — Interpret****

Total = 3 + 7 + 6 + 2 + 1 = 20 ✓. Most common number of siblings is 1.

Common Mistakes

X Confusing 'discrete' and 'continuous' — treating age in years as continuous (it's discrete if measured)

OK Fix: Ask: can it be any value in a range? Height (1.63 m, 1.631 m, ...) is continuous. Shoe size (7, 7.5, 8) is discrete.

X Calling any number-based data 'quantitative' but forgetting that numbers used as labels (e.g., jersey number)

OK Fix: If the numbers are just codes/labels with no meaningful arithmetic (jersey number $\times$ 2 is meaningless), it's qualitative.

X Thinking 'sample' and 'census' are interchangeable — a census covers everyone, a sample covers some

OK Fix: Census = whole population (expensive, time-consuming). Sample = subset (cheaper, faster, but can have bias).

X Tally marks recorded incorrectly — grouping inconsistently or missing the diagonal cross on the groups of 5

OK Fix: Always group by 5: four vertical lines then a diagonal cross through them ($\cancel{\{\}}\{\}$). This makes counting totals easy.

Summary

- Primary data is collected by you; secondary data is from existing sources.
- Qualitative data describes categories; quantitative data involves numbers.
- Discrete data takes only specific values (counts); continuous data can take any value in a range.

(measurements).

- A tally chart organises raw data into groups of 5 for easy frequency counting.
- Choose collection methods carefully — surveys, experiments, observations, census, or sample — based on cost, time, and accuracy needs.

H1_Q2: Frequency Tables

Hook

A factory records the weights of 200 cereal boxes coming off the production line. The weights are not all the same — some are 498 g, some 502 g, some 500.3 g, and so on. Listing all 200 numbers is useless. How do you make sense of this mess? The answer: group the data into intervals and build a frequency table.

Prerequisites

- Understanding of tally charts and frequency (H1_Q1)
- Basic arithmetic (addition, division)
- Concept of inequalities for class boundaries (e.g., $10 \leq x < 20$)

Core Explanation

A frequency table organises raw data by grouping values and counting how many fall into each group.

****Simple Frequency Table****

For discrete data with few distinct values, list each value and count its frequency.

Shoe size	Frequency
6	3
7	8
8	12
9	7
10	2

****Grouped Frequency Table****

When data has many different values (especially continuous data), group them into class intervals.

Weight (g)	Frequency
$495 \leq w < 498$	5
$498 \leq w < 501$	23
$501 \leq w < 504$	16
$504 \leq w < 507$	6

****Class Intervals and Boundaries****

- **Class width** = upper boundary – lower boundary (e.g., $498 - 495 = 3$ g)
- **Upper class boundary** and **lower class boundary** define the exact limits
- Intervals should be: (a) mutually exclusive (no overlap), (b) equal width where possible, (c) cover all data

Midpoints

The midpoint (or class mark) of an interval is the average of its lower and upper boundaries:

$$\text{Midpoint} = \frac{\text{Lower boundary} + \text{Upper boundary}}{2}$$

Midpoints are used when estimating the mean from grouped data.

Cumulative Frequency Table

Shows the running total: how many data points are $\leq$ each upper boundary.

Weight (g)	Frequency	Cumulative Frequency
$495 \leq w < 498$	5	5
$498 \leq w < 501$	23	$5 + 23 = 28$
$501 \leq w < 504$	16	$28 + 16 = 44$
$504 \leq w < 507$	6	$44 + 6 = 50$

The last cumulative frequency equals the total number of data points.

Worked Example

The times (in minutes) for 30 students to complete a puzzle:

12.3, 8.7, 15.1, 10.4, 9.2, 14.8, 11.5, 13.0, 7.9, 16.2,
 9.8, 12.7, 10.1, 14.2, 8.3, 11.9, 13.5, 15.8, 10.6, 12.0,
 14.5, 9.1, 11.2, 13.8, 8.9, 12.5, 10.8, 15.4, 11.1, 14.0

Step 1 — Choose class intervals

Range: 7.9 to 16.2. Use intervals of width 2 minutes.

Time (min)	Tally	Frequency
$7 \leq t < 9$	IIII	4
$9 \leq t < 11$	II	7
$11 \leq t < 13$	III	8
$13 \leq t < 15$	I	6
$15 \leq t < 17$		5

Total = $4 + 7 + 8 + 6 + 5 = 30$ ✓

Step 2 — Find midpoints

Interval	Midpoint formula	Midpoint
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----- ----- -----
$7 \leq t < 9$ $(7+9)/2$ 8
$9 \leq t < 11$ $(9+11)/2$ 10
$11 \leq t < 13$ $(11+13)/2$ 12
$13 \leq t < 15$ $(13+15)/2$ 14
$15 \leq t < 17$ $(15+17)/2$ 16

****Step 3 — Build cumulative frequency table****

Time (min)	Frequency	Cumulative Frequency
-----	-----	-----
7 ≤ t < 9	4	4
9 ≤ t < 11	7	11
11 ≤ t < 13	8	19
13 ≤ t < 15	6	25
15 ≤ t < 17	5	30

Interpretation: 19 students took less than 13 minutes. 11 students took at least 11 minutes but less than 13.

Common Mistakes

X Overlapping class intervals, e.g., 0-10 and 10-20 — where does a value of exactly 10 go?

OK Fix: Use inequalities to make boundaries exclusive: $0 \leq x < 10$ and $10 \leq x < 20$. Or explicitly state inclusive

X Using class intervals of different widths when equal widths work better — makes comparison and

OK Fix: Use equal-width intervals unless the data distribution makes it impractical (very sparse at the extremes)

X Forgetting to include all data values — the lowest and highest values must fall within intervals.

OK Fix: Check the first interval's lower bound $\leq$ minimum value, and the last interval's upper bound $>$ maximum

X Calculating cumulative frequency as the frequency of each row individually instead of the running

OK Fix: Each cumulative frequency = previous cumulative frequency + current row frequency. Always increasing

Summary

- A frequency table organises data by listing values (or intervals) alongside their counts.
- Grouped frequency tables use class intervals — ensure they are equal in width, non-overlapping, and cover all data.
- The midpoint of a class interval = $(\text{lower bound} + \text{upper bound}) / 2$, used for estimating the mean.
- Cumulative frequency tables show running totals (number of values $\leq$ a given upper bound).
- Class intervals should use clear inequality notation to avoid boundary ambiguity.

H1_Q3: Bar Charts and Pictograms

Hook

A school canteen records the number of meals sold each day: Monday 120, Tuesday 95, Wednesday 110, Thursday 130, Friday 150. A table tells you the numbers, but a well-drawn bar chart shows you in an instant that Friday is the busiest and Tuesday the quietest. Pictures and bars turn boring numbers into stories.

Prerequisites

- Understanding of frequency tables (H1_Q2)
- Basic arithmetic (addition, subtraction)
- Reading scales on axes

Core Explanation

Bar charts and pictograms are visual ways to display categorical or discrete data.

****Bar Charts****

- Rectangular bars of equal width represent each category
- The ****height**** (or length, for horizontal bars) is proportional to the frequency
- Bars are separated by gaps (unlike histograms — see H1_Q4)
- Axes must be clearly labelled with a title and scale

****Drawing a bar chart:****

1. Draw and label the x-axis with categories
2. Draw and label the y-axis with frequency (start at 0)
3. Choose a consistent scale (e.g., 1 square = 10 units)
4. Draw each bar to the correct height with equal widths and gaps

****Compound (Stacked) Bar Charts****

Used when data has two or more sub-categories. Each bar is divided into segments showing the contribution of each sub-category.

Example: Favourite sport by gender — each sport category has one bar divided into 'Boys' and 'Girls' sections.

****Pictograms****

- Use symbols or pictures to represent data
- A ****key**** shows the value represented by one symbol
- Each symbol represents a fixed number of items (e.g., $\square$ = 10 apples)
- Half or quarter symbols represent fractional amounts

****Drawing a pictogram:****

1. Choose a symbol and decide what each symbol represents (the key)

2. Draw the appropriate number of symbols for each category (partial symbols allowed)
3. Include the key clearly

****Interpreting from Charts****

- Read bar heights against the scale to find frequencies
- Compare categories directly from bar sizes
- For compound bar charts, compare segments within a bar or across bars
- Estimate values when bars fall between grid lines

Worked Example

The frequency table shows favourite ice cream flavours among 40 students:

Flavour	Frequency
Chocolate	14
Vanilla	10
Strawberry	8
Mint	5
Other	3

****Part A — Bar chart****

- x-axis: flavours (categories)
- y-axis: frequency (scale 0 to 16, increments of 2)
- Draw 5 bars of equal width, separated by gaps, heights: 14, 10, 8, 5, 3
- Label axes and title: 'Favourite Ice Cream Flavours'

****Part B — Pictogram (using = 2 students)****

Flavour	Pictogram
Chocolate	(14 → 7 symbols)
Vanilla	(10 → 5 symbols)
Strawberry	(8 → 4 symbols)
Mint	(4 full) + $\frac{1}{2}$ (5 → $2\frac{1}{2}$ symbols)
Other	(2) + $\frac{1}{2}$ (3 → $1\frac{1}{2}$ symbols)

Key: = 2 students, $\frac{1}{2}$ = 1 student

****Part C — Interpretation****

- Chocolate is the most popular flavour (14 students, 35%)
- 'Other' is the least popular (3 students)
- Chocolate + Vanilla account for 24 out of 40 students (60%)

Common Mistakes

X Not starting the y-axis at zero — this can exaggerate or minimise differences between bars, misl

OK Fix: Always start the frequency axis at 0. If you must start higher (e.g., to show small differences), clearly indicate the scale.

X Making bars unequal in width or omitting gaps between bars in a bar chart (confusing it with a histogram)

OK Fix: Bar charts have gaps between bars because categories are distinct. Histograms have no gaps because they represent continuous data.

X Pictogram symbols not matching the key — e.g., drawing 4 symbols when the key says 1 symbol represents 5 items

OK Fix: Divide the frequency by the key value: $18 \div 5 = 3.6 \rightarrow 3$ full symbols + 0.6 of a symbol (3 full + $\frac{1}{2}$ + $\frac{1}{10}$).

X Forgetting to include a key in a pictogram — the reader cannot interpret the symbols without it.

OK Fix: Always include a clear key: 'Each symbol represents 10 books' or similar. The key is the most important part of a pictogram.

Summary

- Bar charts display categorical/discrete data with equal-width, separated bars whose heights represent frequencies.
- Compound (stacked) bar charts show sub-categories within each main category as stacked segments.
- Pictograms use symbols to represent data; each symbol's value is defined by a key.
- Always start bar chart axes at zero and label both axes clearly.
- When interpreting charts, read bar heights against the scale and compare across categories.
- A pictogram key is essential — without it, the data is meaningless.

H1_Q4: Histograms

Hook

A school measures the heights of all 200 students and groups them: 140–150 cm (34 students), 150–155 cm (62 students), 155–165 cm (80 students), 165–180 cm (24 students). Notice something? The intervals have different widths (10, 5, 10, 15). If you draw a bar chart with heights as frequencies, the 150–155 group looks shorter than the 155–165 group — but that's because it's a narrower range, not because fewer people are in that height zone per cm. Histograms fix this distortion.

Prerequisites

- Understanding of grouped frequency tables (H1_Q2)
- Concept of class intervals and class width
- Bar charts (H1_Q3) — knowing how histograms differ
- Basic arithmetic (division, multiplication)

Core Explanation

A histogram is a graphical representation of a grouped frequency distribution for **continuous** data. It shows the distribution of data over continuous intervals.

Histograms vs Bar Charts — Key Differences

Feature	Bar Chart	Histogram
Data type	Categorical / Discrete	Continuous
Bar spacing	Gaps between bars	No gaps between bars
Bar width	All equal (fixed)	Varies if class intervals differ
Vertical axis	Frequency	Frequency density (usually)
Bar area	Not meaningful	Proportional to frequency

Frequency Density

When class intervals have **unequal widths**, you cannot use frequency on the y-axis — wider intervals would artificially look more important. Instead, plot **frequency density**:

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$$

This ensures the **area of each bar** is proportional to the frequency.

$$\text{Area of bar} = \text{Frequency Density} \times \text{Class Width} = \text{Frequency}$$

Drawing a Histogram

1. Calculate class width for each interval (upper – lower boundary)
2. Calculate frequency density for each interval
3. Draw continuous x-axis with the data scale (no gaps between bars)
4. Draw y-axis for frequency density
5. Draw bars with heights = frequency density, widths = class width

Interpreting a Histogram

- The **area** of each bar tells you the frequency (not the height)
- Taller bars indicate higher density (more data per unit), not necessarily more data overall
- To find frequency from a histogram: Frequency = Frequency Density × Class Width

Worked Example

The table shows the distribution of journey times (in minutes) for 200 commuters:

Time (minutes)	Frequency	Class Width	Frequency Density
$0 \leq t < 10$	20	10	$20 \div 10 = 2.0$
$10 \leq t < 20$	60	10	$60 \div 10 = 6.0$
$20 \leq t < 30$	80	10	$80 \div 10 = 8.0$
$30 \leq t < 40$	30	10	$30 \div 10 = 3.0$
$40 \leq t < 60$	10	20	$10 \div 20 = 0.5$

Step 1 — Notice the unequal interval

The last interval is 40–60 (width 20), double the others. If we plotted frequency (10) directly, it would look like a small bar — but the actual density is even lower: only 0.5 people per minute.

Step 2 — Draw the histogram

- x-axis: Time (minutes), continuous scale 0 to 60
- y-axis: Frequency density, scale 0 to 10
- Draw 5 bars with no gaps:
 - 0–10: height 2.0, width 10
 - 10–20: height 6.0, width 10
 - 20–30: height 8.0, width 10
 - 30–40: height 3.0, width 10
 - 40–60: height 0.5, width 20

****Step 3 — Verify areas****

- Area of 0–10 bar = $2.0 \times 10 = 20$ ✓
- Area of 40–60 bar = $0.5 \times 20 = 10$ ✓

****Step 4 — Interpretation****

- Most journeys are 20–30 minutes (highest frequency density of 8.0)
- The 40–60 interval has the longest range but the lowest density — very few long journeys
- Total area = $20 + 60 + 80 + 30 + 10 = 200$ ✓

Common Mistakes

X Putting 'Frequency' on the y-axis instead of 'Frequency Density' when intervals are unequal — th

OK Fix: Always calculate and plot frequency density = frequency $\div$ class width when class widths vary. The are

X Leaving gaps between bars in a histogram as if it were a bar chart — this incorrectly implies the

OK Fix: Histograms have no gaps between bars because the data is continuous — the next interval starts exact

X Confusing class width with the difference between the recorded values rather than the boundary

OK Fix: For continuous data, class boundaries may differ from the stated limits. Be precise: if the interval is 10

X Thinking the tallest bar in a histogram always represents the most data — but if intervals are un

OK Fix: Compare areas (height $\times$ width), not heights alone. The frequency = area of the bar. A short wide bar c

Summary

- A histogram displays continuous grouped data with no gaps between bars.
- Unlike bar charts, bar width can vary when class intervals are unequal — the area represents the frequency.
- Frequency density = frequency $\div$ class width; this is the standard y-axis variable for histograms with unequal intervals.
- Area of a bar = frequency density $\times$ class width = frequency of that interval.
- Always verify that total area equals total number of data points.
- The tallest bar has the highest density, but not necessarily the highest frequency — check the area.

H1_Q5: Mean, Median and Mode

Hook

A survey of 30 students asked, 'How many books did you read last month?' The raw replies are: 2, 0, 3, 1, 5, 2, 4, 0, 1, 3, 2, 2, 6, 1, 3, 4, 2, 0, 1, 2, 3, 5, 1, 2, 4, 3, 0, 2, 1, 3. Which 'average' best represents this group? The mean might be pulled by the few heavy readers, the median hides the extremes, and the mode tells you the most common experience. Each measure tells a different story about the same data.

Prerequisites

- Ability to add, subtract, multiply and divide whole numbers and decimals
- Understanding of ordered lists and sorting numbers in ascending order
- Familiarity with frequency tables (tally charts)

Core Explanation

The three main measures of central tendency describe the 'centre' of a data set in different ways.

Mean — the arithmetic average. Calculated by summing all values and dividing by the number of values. For raw data: $\text{mean} = (\sum x) / n$. For a frequency table: $\text{mean} = \sum(f \cdot x) / \sum f$, where f is the frequency of each value x .

Median — the middle value when data is arranged in order. For n values, the median is at position $(n+1)/2$. If n is odd, take that value directly. If n is even, take the mean of the two middle values. For grouped data (intervals), the median lies in the interval where the cumulative frequency first reaches or exceeds $n/2$.

Mode — the most frequently occurring value. A data set may have one mode (unimodal), more than one (bimodal/multimodal), or no mode (if every value appears once). For grouped data, the modal class is the interval with the highest frequency.

When comparing data sets, always consider which measure is most appropriate. The mean uses every data point but is sensitive to outliers. The median is resistant to outliers but ignores extreme values. The mode is useful for categorical or discrete data but may not exist or be meaningful.

Worked Example

A class of 8 students scored the following marks on a test: 12, 7, 15, 9, 12, 10, 14, 7.

Step 1: Mean

$$\text{Sum} = 12 + 7 + 15 + 9 + 12 + 10 + 14 + 7 = 86$$

$$n = 8$$

$$\text{Mean} = 86 \div 8 = 10.75$$

Step 2: Median

Arrange in order: 7, 7, 9, 10, 12, 12, 14, 15

$n = 8$ (even), so median = mean of 4th and 5th values = $(10 + 12) \div 2 = 11$

****Step 3: Mode****

7 appears twice, 12 appears twice, all other values appear once.

The data set has two modes: 7 and 12 (bimodal).

****Now with a frequency table:****

A survey asked 20 people how many cups of tea they drink per day.

Cups (x)	0	1	2	3	4
Frequency (f)	3	7	5	3	2

Mean = $\Sigma(f \cdot x) \div \Sigma f = (0 \times 3 + 1 \times 7 + 2 \times 5 + 3 \times 3 + 4 \times 2) \div 20 = (0 + 7 + 10 + 9 + 8) \div 20 = 34 \div 20 = 1.7$ cups

Median position = $(20+1)/2 = 10.5$ — find the 10th and 11th values. Cumulative frequencies: 3, 10, 15, 18, 20. The 10th value is in the '1 cup' group (cumulative freq hits 10), and the 11th is also in '1 cup' (cumulative freq 15). So both middle values = 1 $\rightarrow$ median = 1.

Mode = 1 cup (highest frequency = 7).

Common Mistakes

X Forgetting to sum the frequencies (Σf) when calculating the mean from a frequency table — student error

OK Fix: Always divide $\Sigma(f \cdot x)$ by Σf , not by the number of rows. Each row represents many data points, so the denominator must be the total frequency.

X Taking the median as the $(n+1)/2$ -th value rounded up, even when n is even — this incorrectly picks a single value

OK Fix: When n is even, the median is the mean of the two middle numbers at positions $n/2$ and $(n/2 + 1)$. Always average the two middle values.

X Confusing 'no mode' with 'mode = 0'. If no value repeats, there is no mode — writing '0' implies a mode of 0

OK Fix: State 'no mode' when each value occurs exactly once. If the value 0 appears most frequently, then the mode is 0.

Summary

- Mean = sum of all values $\div$ number of values; it uses all data but is affected by outliers.
- Median = middle value after sorting; resistant to outliers, best for skewed distributions.
- Mode = most frequent value; can be used for non-numerical data; may be unimodal, bimodal, or non-existent.
- For frequency tables: mean uses $\Sigma(f \cdot x)/\Sigma f$, median uses cumulative frequency, mode is the value with highest frequency.
- Always consider context when choosing which measure of central tendency to report.

H1_Q6: Range, Quartiles and Box-and-Whisker Plots

Hook

Teacher A says her class had a mean score of 72%. Teacher B's class also had a mean of 72%. Are the classes equally prepared? Not necessarily — one class might have scores tightly clustered between 65%

and 79%, while the other ranges from 20% to 100% with most students bunched at the extremes. Measures of spread — range, quartiles, and the interquartile range — reveal what the average hides.

Prerequisites

- Understanding of median as a measure of central tendency (H1_Q5)
- Ability to sort numerical data in ascending order
- Basic fraction and decimal arithmetic

Core Explanation

Measures of spread (dispersion) describe how varied or spread out a data set is.

Range — the simplest measure: maximum value – minimum value. It is sensitive to outliers because only the two extreme values are used.

Quartiles — divide the sorted data into four equal parts:

- Q1 (lower quartile): median of the lower half of data (25th percentile)
- Q2 (median): the middle value (50th percentile)
- Q3 (upper quartile): median of the upper half of data (75th percentile)

To find quartiles:

1. Sort data in ascending order.
2. Find the median (Q2).
3. Q1 is the median of the values below Q2 (excluding Q2 if n is odd).
4. Q3 is the median of the values above Q2.

Interquartile Range (IQR) = $Q3 - Q1$. This measures the spread of the middle 50% of the data. Unlike the range, the IQR is not affected by outliers.

Box-and-whisker plot (box plot): A visual display showing five key statistics:

- Minimum value (lower whisker end)
- Q1 (left edge of box)
- Median (line inside box)
- Q3 (right edge of box)
- Maximum value (upper whisker end)

The box represents the middle 50% of data; each whisker represents the bottom and top 25%.

Percentiles — values below which a given percentage of data falls. Q1 = 25th percentile, Median = 50th percentile, Q3 = 75th percentile. The k-th percentile can be found by: $\text{position} = (k/100) \times (n + 1)$.

Worked Example

The ages (in years) of 11 people at a club meeting are: 18, 22, 25, 27, 29, 31, 34, 38, 42, 45, 60.

Step 1: Range

Maximum = 60, Minimum = 18

Range = $60 - 18 = 42$ years

****Step 2: Quartiles****

Data is already sorted ($n = 11$).

Q2 (median) = position $(11+1)/2 = 6$ th value $\rightarrow 31$

Lower half: 18, 22, 25, 27, 29 (5 values). Q1 = 3rd value = 25.

Upper half: 34, 38, 42, 45, 60 (5 values). Q3 = 3rd value = 42.

IQR = $Q3 - Q1 = 42 - 25 = 17$ years.

****Step 3: Box-and-whisker plot****

Five-number summary: Min = 18, Q1 = 25, Med = 31, Q3 = 42, Max = 60.

Draw a number line from about 15 to 65. Draw a box from Q1 (25) to Q3 (42) with a vertical line at the median (31). Extend whiskers from the box edges to the minimum (18) and maximum (60).

****Using percentiles****

Find the 60th percentile.

Position = $(60/100) \times (11 + 1) = 0.6 \times 12 = 7.2$

This is between the 7th value (34) and 8th value (38).

60th percentile = $34 + 0.2 \times (38 - 34) = 34 + 0.8 = 34.8$ years.

Common Mistakes

X Including the median (Q2) in both halves when finding Q1 and Q3 for data sets with an odd number of values

OK Fix: When n is odd, exclude the median itself from both the lower and upper halves before finding Q1 and Q3.

X Drawing a box-and-whisker plot with the whiskers going from min-to-Q1 and Q3-to-max inside the box

OK Fix: The whiskers extend outward from the box edges to the extreme values. The box itself spans Q1 to Q3.

X Confusing the range with the interquartile range — using range to describe the 'middle half' of data

OK Fix: Range = $\text{Max} - \text{Min}$ (covers all data). IQR = $Q3 - Q1$ (covers only the middle 50%). Use IQR when you want to describe the spread of the middle half of the data.

Summary

- Range = largest value – smallest value; simple but outlier-sensitive.
- Quartiles divide sorted data into four equal parts: Q1 (25th percentile), Q2/median (50th), Q3 (75th).
- Interquartile Range (IQR) = $Q3 - Q1$; measures spread of the middle 50% of data; robust to outliers.
- Box-and-whisker plots visualise the five-number summary: Min, Q1, Median, Q3, Max.
- Percentiles generalise the idea: the k -th percentile is the value below which $k\%$ of data falls.

H1_Q7: Scatter Diagrams and Correlation

Hook

Does studying more hours guarantee higher exam scores? A scatter diagram of 15 students' study hours vs. exam scores might show a clear upward trend — but one student who studied 40 hours and scored only 30% tells a different story. Scatter diagrams let us see relationships (or the lack of them) at a glance, before we jump to conclusions about cause and effect.

Prerequisites

- Plotting points on a Cartesian coordinate grid
- Drawing straight lines on a graph
- Understanding of 'mean' and 'median' (H1_Q5)

Core Explanation

A **scatter diagram** (scatter plot) displays paired numerical data as points on a coordinate grid, with one variable on the x-axis and the other on the y-axis.

Correlation describes the strength and direction of a linear relationship:

- **Positive correlation**: as x increases, y tends to increase. Points cluster around an upward-sloping line.
- **Negative correlation**: as x increases, y tends to decrease. Points cluster around a downward-sloping line.
- **No correlation**: no clear pattern; points are scattered randomly.

Correlation strength can be described as:

- **Strong**: points lie close to a straight line
- **Weak**: points are loosely scattered around a line
- **Perfect**: all points lie exactly on a straight line (rare in real data)

Line of best fit: A straight line drawn through a scatter plot that best represents the trend. It should pass through the mean point ($\bar{x}$, $\bar{y}$) and have roughly equal numbers of points above and below it. The line is used to estimate values.

Interpolation: Estimating a y-value for an x-value that lies within the range of the data. Generally reliable.

Extrapolation: Estimating a y-value for an x-value beyond the range of the data. Less reliable because the trend may change outside the observed data.

Outliers: Points that do not follow the general trend. They can be identified as points far from the line of best fit. Investigate outliers — they may indicate measurement error or a special case.

⚠ **Correlation does not imply causation.** Even a strong correlation does not prove that one variable causes the other. There may be a third (confounding) variable, or the relationship may be coincidental.

Worked Example

Ten students recorded their weekly revision hours and test scores (out of 100):

Hours (x)	2	4	5	6	8	9	10	12	14	15	
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Score (y)	30	42	45	55	60	65	68	78	85	90	

****Step 1: Draw the scatter diagram****

Plot each (x, y) pair on a grid with x-axis labelled 'Revision Hours' (0-16) and y-axis 'Test Score' (0-100). The points show an upward trend from bottom-left to top-right.

****Step 2: Describe the correlation****

The points cluster closely around an imaginary upward line — strong positive correlation.

****Step 3: Draw the line of best fit****

Calculate the mean point: $\bar{x} = (2+4+5+6+8+9+10+12+14+15)/10 = 85/10 = 8.5$ hours. $\bar{y} = (30+42+45+55+60+65+68+78+85+90)/10 = 618/10 = 61.8$. Mean point = (8.5, 61.8).

Draw a straight line through (8.5, 61.8) that balances the points (about 5 above, 5 below).

****Step 4: Interpolation****

Estimate the score for a student who revises 7 hours. From the line: at $x = 7$, read $y \approx 50$. A student with 7 hours would be expected to score about 50.

****Step 5: Extrapolation (with caution)****

Estimate the score for 18 hours. Extend the line: at $x = 18$, read $y \approx 100$ (beyond data range). This suggests a perfect score, but the trend may not hold — students with 18 hours might experience burnout or diminishing returns.

****Outlier example****: If a student with 10 hours scored only 30, that point would lie far below the trend line. Possible reasons: the student didn't actually study effectively, was unwell, or the score was recorded incorrectly.

Common Mistakes

X Claiming causation from correlation alone (e.g., 'ice cream sales cause drowning because both increase in summer')

OK Fix: Correlation only shows a relationship. A third factor (e.g., hot weather) may cause both variables to change.

X Drawing a line of best fit that does not pass through the mean point ($\bar{x}$, $\bar{y}$), or connecting the first and last data points.

OK Fix: The line of best fit must pass through ($\bar{x}$, $\bar{y}$) and have roughly equal numbers of points on each side. It is not necessarily a line connecting the first and last points.

X Extrapolating far beyond the data range and treating the estimate as equally reliable as interpolation.

OK Fix: Extrapolation assumes the trend continues unchanged, which may be false. Always state 'this estimate is based on extrapolation'.

Summary

- Scatter diagrams plot paired data to reveal relationships between two variables.
- Correlation can be positive, negative, or none; strength is described as strong, weak, or perfect.

- The line of best fit passes through the mean point and represents the overall trend.
- Interpolation (estimating within the data range) is reliable; extrapolation (beyond the range) is less reliable.
- Outliers are points that deviate from the trend — investigate rather than ignore them.
- Correlation does not equal causation — always consider confounding variables.